

SchoolHub SA — Complete Functional Capabilities Catalogue

Cyber Developers · SchoolHub SA

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Source: commit `ca8089b`, 9 October 2026

Companion: [feature-verification-matrix.md](#), [module-inventory.md](#)

This catalogue lists what the software implements. Status words are defined in [README.md](#). Items under “Planned or unavailable” must not be sold as working features.

1. Platform shape

- Multi-institution data. Each operational row is scoped to a school. Super Admin can cross institutions. Other roles cannot.
- Twelve roles: Super Admin, School Admin, Principal, Teacher/Lecturer, Learner, Parent/Guardian, Finance Officer, Admissions Officer, HR Officer, Staff, Security.
- Per-user permission grants and denies on top of the role.
- Modules can be disabled per school. A disabled module hides related permissions for non-Super Admin users.
- Terminology switches between school language and college language.
- Institution types: school, primary, high, combined, college, TVET, training centre, training institution.
- Curricula: CAPS, NSC, TVET/NQF, custom.
- Calendar structures: four terms, two semesters, custom.
- Time zone Africa/Johannesburg. Money in ZAR.
- Brand assets served from `/brand/logo.png` and `/brand/mark.png`.

Authentication. Email and password, session version for forced sign-out, invitation with temporary password, forced reset, forgot-password link, admin set-password for Super Admin and School Admin. School Admin cannot set a Super Admin password.

Licence. Signed licence check against the licence server. Statuses trial, active, grace, expired, suspended, revoked. Grace is 14 days. Heartbeat is automatic. Feature flags can turn commercial modules off.

2. Module catalogue

Institution and identity

Capability	Status	Notes
Admin dashboard	Implemented, not fully tested as a single production script	<code>/admin/dashboard</code>
School profile, province list, branding	Implemented and tested in settings tests	<code>/admin/settings</code>
Campuses	Implemented, not fully tested	No separate “departments” entity. Department is text on staff.
Academic sessions, terms, rollover	Implemented, not fully tested end to end	Rollover is staff-run
Grades, classes, subjects, courses, modules	Implemented	College labels differ
Users, invitations, role defaults	Implemented and tested in permission and mail tests	Delivery needs Resend or school SendGrid
Per-user grants and denies	Implemented and tested	<code>/admin/users/{id}/permissions</code>
Admin set password	Implemented and tested locally	On master via #72 . Not a substitute for the invitation flow
Super Admin institutions	Implemented	Vendor only

Admissions

Capability	Status	Notes
Public application form and status tracker	Implemented	<code>/apply , /{slug}/apply , /apply/status</code>
Staff pipeline with the statuses in the user manual	Implemented	Interview required is a status, not a scheduler
Application documents	Implemented	

Capability	Status	Notes
Deposit invoice, paid flag, waiver	Implemented	Waiver needs admissions.deposit_waive
Create learner and enrolment on accept	Implemented	Grade or course name must match
Offer letters	Implemented as letters desk	Not auto-sent on status change
Interviews as appointments	Planned or unavailable	Status flag only
Automatic acceptance	Planned or unavailable	Staff set every status

Learner records

Capability	Status	Notes
Registration, profile, guardians, documents	Implemented	
Statuses applicant through withdrawn	Implemented	
SA ID encryption and sensitive-field permission	Implemented	
Change log	Implemented	StudentChangeLog
Portal user link	Implemented, separate step	Learner row can exist without a login
Transfers	Partially implemented	Letters, status, and promotion outcome. No standalone transfer wizard that moves a learner between schools by itself

Teaching

Capability	Status	Notes
Timetable with room, educator, meeting URL	Implemented	Join link is an external URL
Lesson plans and curriculum topics	Implemented	Completion is staff-updated
Class and session attendance	Implemented and tested in attendance areas	

Capability	Status	Notes
Gate writes daily learner attendance	Implemented and covered by gate tests	Does not write period registers
Assessments and weighted marks	Implemented	Types listed in the user manual
Homework and submissions	Implemented	
Online exam attempt (MCQ, true/false, short answer)	Implemented	No proctoring
Report card PDF, skip rules, family notice	Implemented	Staff must issue
Certificates	Implemented	
Promotion rules and outcomes	Implemented	Staff confirm
Educator reviews by learners	Implemented	Licence feature defaults on
Hosted virtual classroom	Planned or unavailable	External URL only
AI marking or AI tutor	Planned or unavailable	ai_features defaults off and has no portal

Finance

Capability	Status	Notes
Fee schedule, structures, charges, instalments	Implemented	
Invoices and PDFs	Implemented	Draft and cancelled reject collections
Manual collection, bank-reference duplicate check	Implemented	
Pending, verified, approved, rejected	Implemented	
Ledger, allocations, credit notes, aid	Implemented	
Debtors and age analysis	Implemented	On demand
Budgets and projects	Implemented on admin finance	
Expenses, suppliers, income, accounts	Implemented	

Capability	Status	Notes
PayFast, Ozow, Yoco, Paystack settings	Requires configuration	Paystack is the current replacement path for the older PayPal method name
Gateway idempotency on reference	Implemented	
Receipt numbering after a normal first receipt	Implemented	
Reversal of a referenced payment	Partially implemented	Defect. Not production-ready
Receipt number after -REV or -R	Partially implemented	Defect. Not production-ready
Concurrent approval lock	Partially implemented	Status check is not row-locked
Automatic fee reminders	Partially implemented	Rules and manual send exist. Daily job held
Automatic recurring expenses	Partially implemented	Definition exists. Daily job held

People

Capability	Status	Notes
Employee file, contract, documents, salary structure	Implemented	
Staff attendance including remote and on-leave	Implemented	Gate can stamp check-in and check-out
Leave requests, policies, entitlements	Implemented	
Monthly leave accrual job	Partially implemented	Route exists. Scheduler held
Timesheets	Implemented	
Payroll prepare, approve, finalise, payslip PDF	Implemented	Finalise can post to finance once
Biometric clock import	Planned or unavailable	Explicitly not implemented

Communication and website

Capability	Status	Notes
Announcements with audiences	Implemented	

Capability	Status	Notes
Internal messages	Implemented	Security has no mailbox
In-app notifications	Implemented	
Platform email via Resend	Requires configuration	Acceptance is not inbox delivery. No webhook
School SendGrid	Requires configuration	Used only when enabled, keyed, and From is deliverable
SMS via Twilio	Requires configuration	Licence feature defaults off
Bulk fee email/SMS	Requires configuration	Manual
Public website, gallery, FAQ, admissions copy	Implemented	
WhatsApp click-to-chat on the public site	Requires configuration	Number on the school profile. No inbox
WhatsApp messaging product	Planned or unavailable	Flag defaults off

Campus access

Capability	Status	Notes
Gates, policy, day boundary, duplicate window	Implemented and tested	
Cards for learners and staff, deactivate	Implemented	Admin issue. Security scans
QR, barcode, camera, manual scan	Implemented and tested	
RFID/NFC as a method name resolved to a person	Partially implemented	No device fleet certified in this review
Biometric capture	Planned or unavailable	Request is rejected
Visitors book, expected visitors, sign-out	Implemented	
On site and missing checkout	Implemented and tested	Prior-day IN is missing, not still on site

Capability	Status	Notes
Early departure reasons	Implemented	Parent approval workflow for early release is reserved, not the security desk's current path
Gate incidents list	Implemented, not fully tested as a case system	
Security role isolation	Implemented in middleware and tests	Live college account first login not completed
Parent SMS on gate scan	Planned or unavailable	Not part of the scan

Platform operations

Capability	Status	Notes
Audit log	Implemented	
Compliance exports: SA-SAMS package, LURITS promotion, CEMIS marks	Implemented, not fully tested against a department portal	Staff-run
SA-SAMS staging, map, execute, rollback	Implemented	
Import staging cleanup job	Partially implemented	Held
Backup create, download, restore	Implemented	Restore is destructive
Scheduled backups	Partially implemented	Held
Licence heartbeat	Implemented and tested	Automatic, due-only
System health job panel	Implemented	Super Admin
POPIA field controls and redacted mail logs	Implemented	Not a compliance certificate
Library catalogue	Planned or unavailable	Flag defaults off
Public API for third parties	Planned or unavailable	api_access defaults off
Advanced analytics product	Planned or unavailable	Flag defaults off

3. Automation, stated exactly

Chain	What the software does without a person clicking	What still needs a person
Application to learner	On Accepted or Enrolled, creates or links the learner and enrolls them when names match and the deposit rule allows	Every status before that, including interview and offer
Enrolment to class	Stores the enrolment	Class placement and timetable slots
Timetable to learner	Shows slots and an external meeting link	Building the timetable
Marks to report	Calculates averages when staff issue the card	Issuing the card
Invoice to gateway receipt	After the provider confirms, locks the invoice, records the payment, updates the balance, allocates, and writes the ledger	Gateway keys, and the payer completing the hosted page
Manual receipt	Allocates and writes the ledger when the payment is approved or captured as approved	The cashier, and verification for pending methods
Balance to debtor list	Ageing reads current balances	Opening the report, and any reminder send
Gate to daily attendance	Recorded learner IN/OUT updates the daily row in the same transaction	Card issue, and any period register
Gate to staff time	Recorded staff movement can set check-in and check-out	Payroll
Employee to pay	Nothing	Prepare, approve, finalise
Clock to backups	Nothing automatic	Backup screen, or a future reviewed scheduler
Clock to fee reminders	Nothing automatic	Manual send
Clock to licence	Heartbeat when due	Licence server availability

4. Integrations

Integration	Direction	Status
Resend	Platform transactional email	Requires configuration. Official <code>resend</code> package

Integration	Direction	Status
SendGrid	Optional per-school email	Requires configuration. Disabled school config does not block Resend
Twilio	SMS	Requires configuration and licence
Paystack, PayFast, Ozow, Yoco	Hosted fee collection	Requires configuration per school
Licence server	Signed licence check	Required for ongoing verification
SA-SAMS file	Import and export package	Staff-run
LURITS / CEMIS	Export files	Staff-run
WhatsApp	Public click-to-chat link	Optional number
Biometric terminals	—	Unavailable
Resend webhooks	Delivery events	Unavailable

Secrets are encrypted on the school integration record. Screens return configured, sender validity, and provider name. They must not display API keys. This catalogue does not contain any key.

5. Background work

In-process scheduler, started from `instrumentation.ts` on the Node.js runtime, without blocking server startup. A database lease (`scheduler_job_states`) stops two replicas doing the same job. Lease failure is logged and does not crash the process.

Automatic: `license-heartbeat` at 15:00 UTC.

Held: `backups`, `fee-reminders`, `import-cleanup`, `leave-accrual`, `recurring-expenses`.

Protected HTTP routes under `/api/cron/*` remain for an operator who presents the cron secret. Query-string secrets are not accepted.

6. Generated documents

Invoice PDF, receipt PDF, report card PDF, certificate PDF, official letter PDF, payslip PDF, identity card, compliance export files, backup archive. All are on demand except where a payment post

or report issue creates the file as part of that action.

7. API surface

212 route modules live under `src/app/api`. The file list is [appendix-api-routes.txt](#). 213 pages are listed in [appendix-pages.txt](#). Those lists were generated from the source tree at this commit. They are the inventory. They are not a public API contract. `api_access` is off by default and these routes are the application's own UI backend.

8. Tests that support this catalogue

Automated tests cover permissions, licensing, the public licence check, mail provider selection, the scheduler lease and due-only heartbeat, gate behaviour, school branding, and settings, among other suites. A full run during the audit passed 437 tests and failed 1: `tests/teacher-classes.test.ts` still expects the educator groups "Classwork" for attendance and assessments, while the menu uses "Daily". The menu is the product.

Payment reversal and receipt regeneration after a suffix do not have tests that would catch the defects in [known-defects-for-remediation.md](#). That absence is why those features are not marked production-ready.